

Large intestine

Department of Anatomy

The abdomen (6)

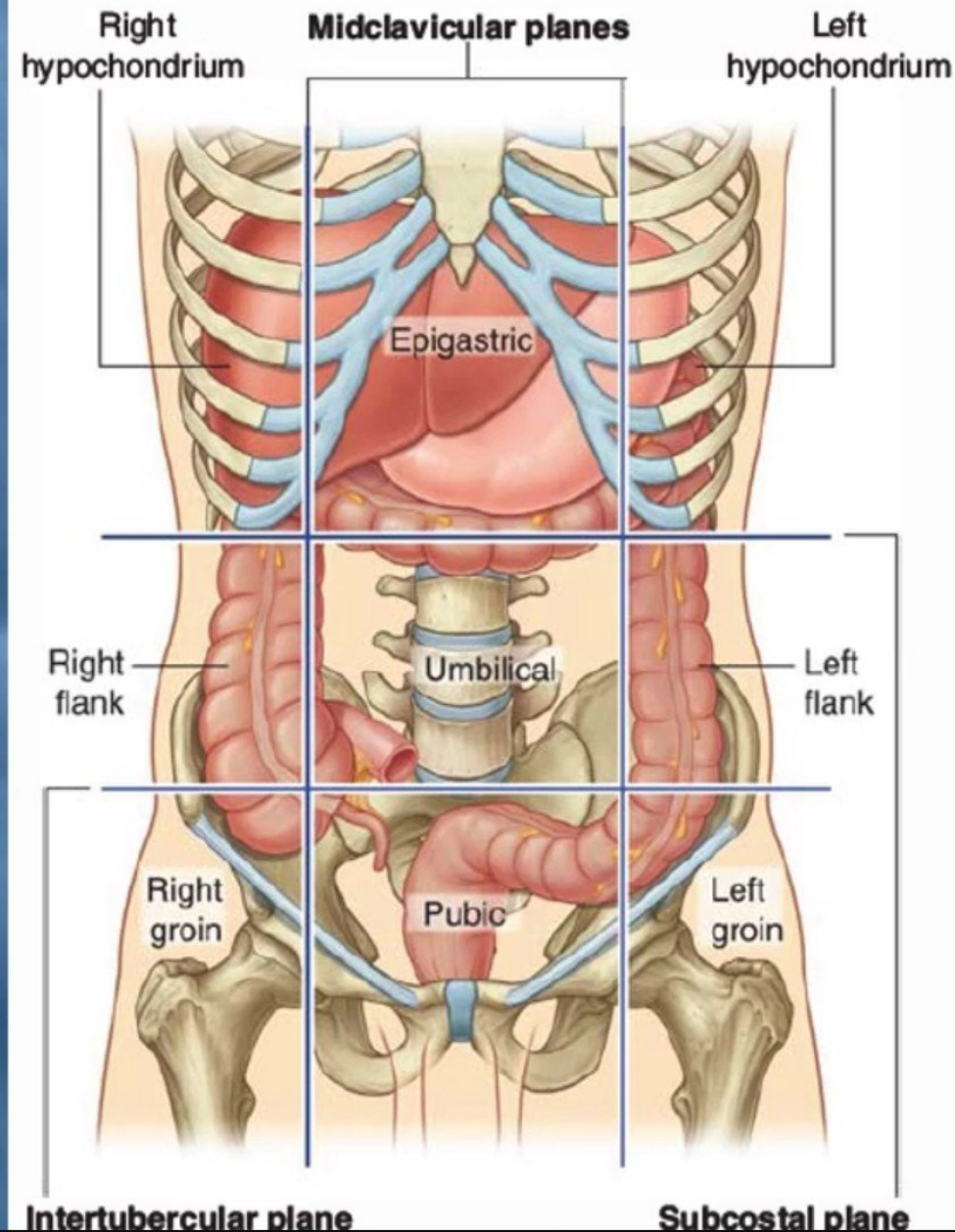
First Part

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Lecture points

- **Position of large intestine related to clinical nine regions pattern.**
- **Functions of the Large Intestine.**
- **Divisions and special features of Large Intestine.**
- **Cecum:**
 - **Location and Description.**
 - **Cecum Relations.**
 - **Arterial supply.**
 - **Venous drainage.**
 - **Lymphatic drainage.**
 - **Ileocecal Valve.**
- **Appendix**
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 - ✓ **Venous drainage.**
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- **Colon:**
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 - ✓ **Location and Description.**
 - ✓ **Relation.**
 - ✓ **Arterial supply.**
 - ✓ **Venous drainage.**
 - ✓ **Lymphatic drainage.**
 - ✓ **Nerve supply.**
- **Transverse Colon**
 - ✓ **Location and Description.**
 - ✓ **Relation.**
 - ✓ **Arterial supply.**
 - ✓ **Venous drainage.**
 - ✓ **Lymphatic drainage.**
 - ✓ **Nerve supply.**
- **Descending colon:**
 - ✓ **Location and Description.**
 - ✓ **Relation.**
 - ✓ **Arterial supply.**
 - ✓ **Venous drainage.**
 - ✓ **Lymphatic drainage.**
 - ✓ **Nerve supply.**
- **Sigmoid colon.**
 - **Arterial supply of GUT.**
 - **Enteric Nervous System.**

Position of large intestine related to clinical nine regions pattern



Functions of the Large Intestine

- The large intestine extends from **the ileum** to **the anus**. (1.5 m long)
- The **rectum** and **anal canal** are considered in the sections on **the pelvis and perineum**.

Main functions:

- **Absorb water** and **electrolytes**.
- The storage of **undigested material until** it can be expelled from the body as feces (**Mass peristaltic movements force feces toward the rectum**).
- **Small amount of nutrients digestion by bacteria**.
 - ✓ The colon absorbs vitamins which are created by the colonic bacteria

Divisions and special features of Large Intestine

Subdivided into:

Cecum, vermiform appendix, colon, rectum, anal canal.

Special features of large intestine:

➤ **Teniae coli:**

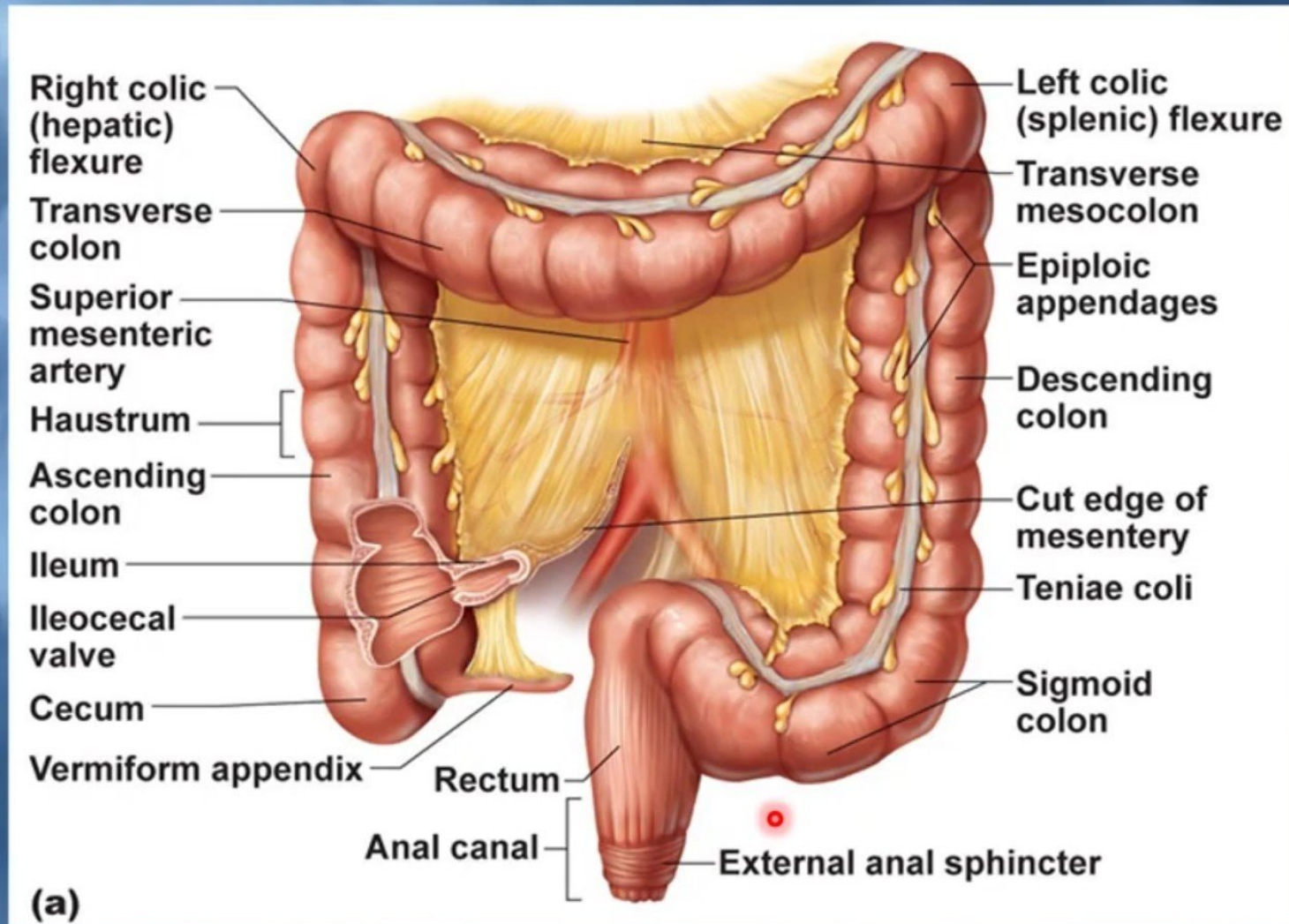
- Thickening of longitudinal muscularis.

➤ **Haustra:**

- Puckering created by teniae coli.

➤ **Epiploic appendages:**

- Fat-filled pouches of visceral peritoneum.



Divisions and special features of Large Intestine

Cecum

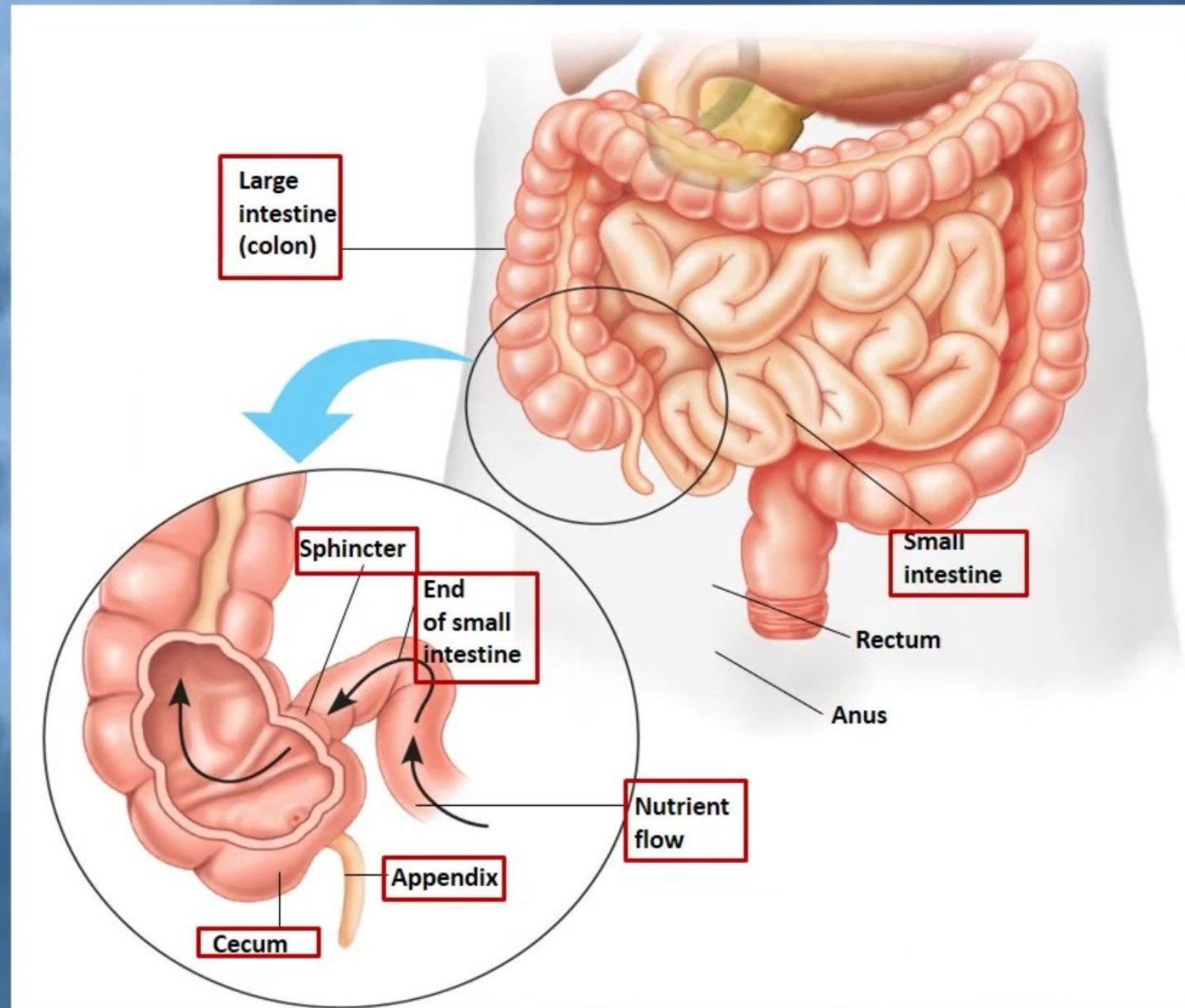
- Blind pouch
- Beginning of large intestine

Vermiform appendix

- Contains lymphoid tissue
- Neutralizes pathogens

Colon

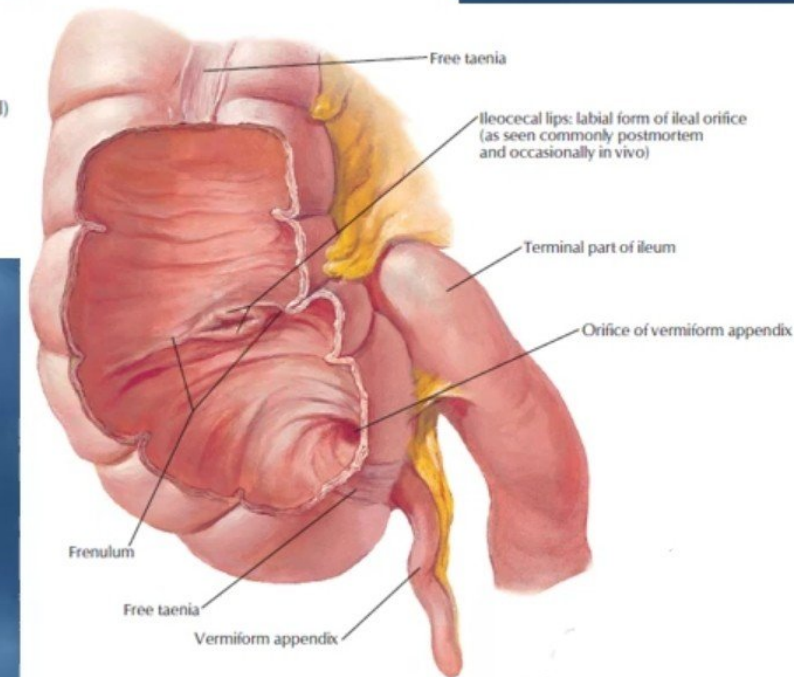
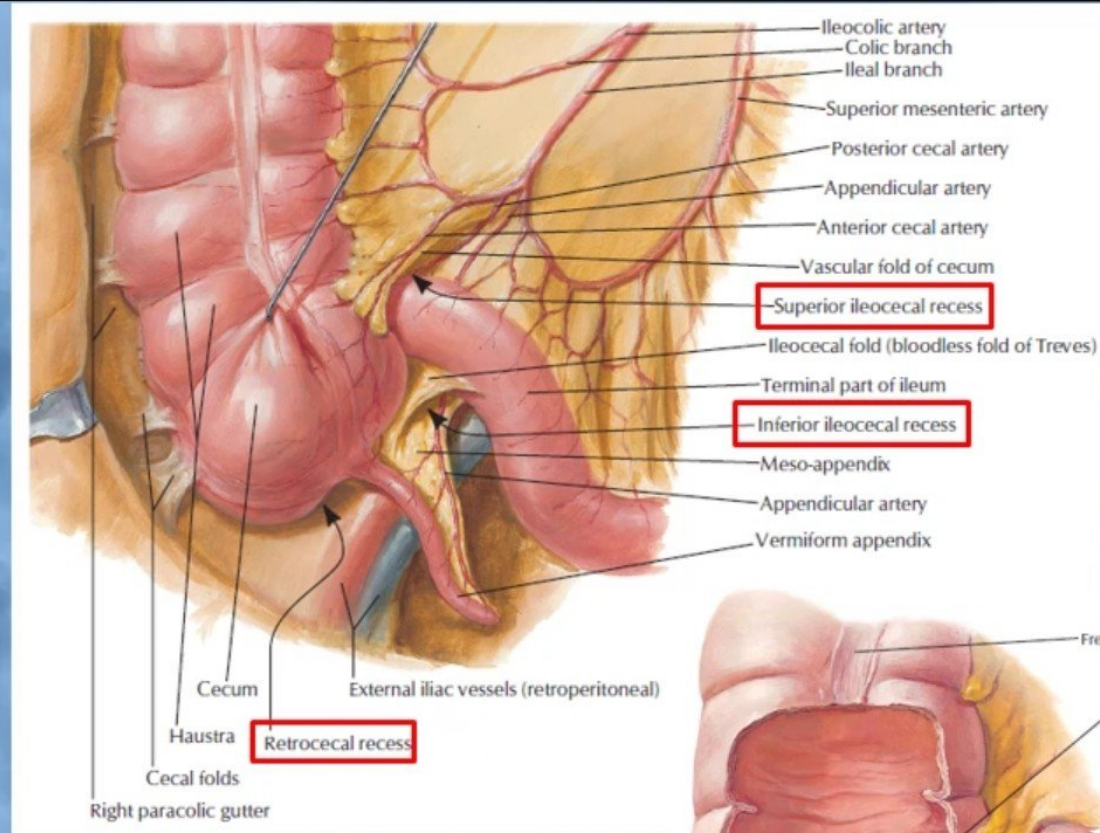
- Divided into distinct segments
 - Ascending, transverse, descending, and sigmoid colon.



Cecum

Location and Description

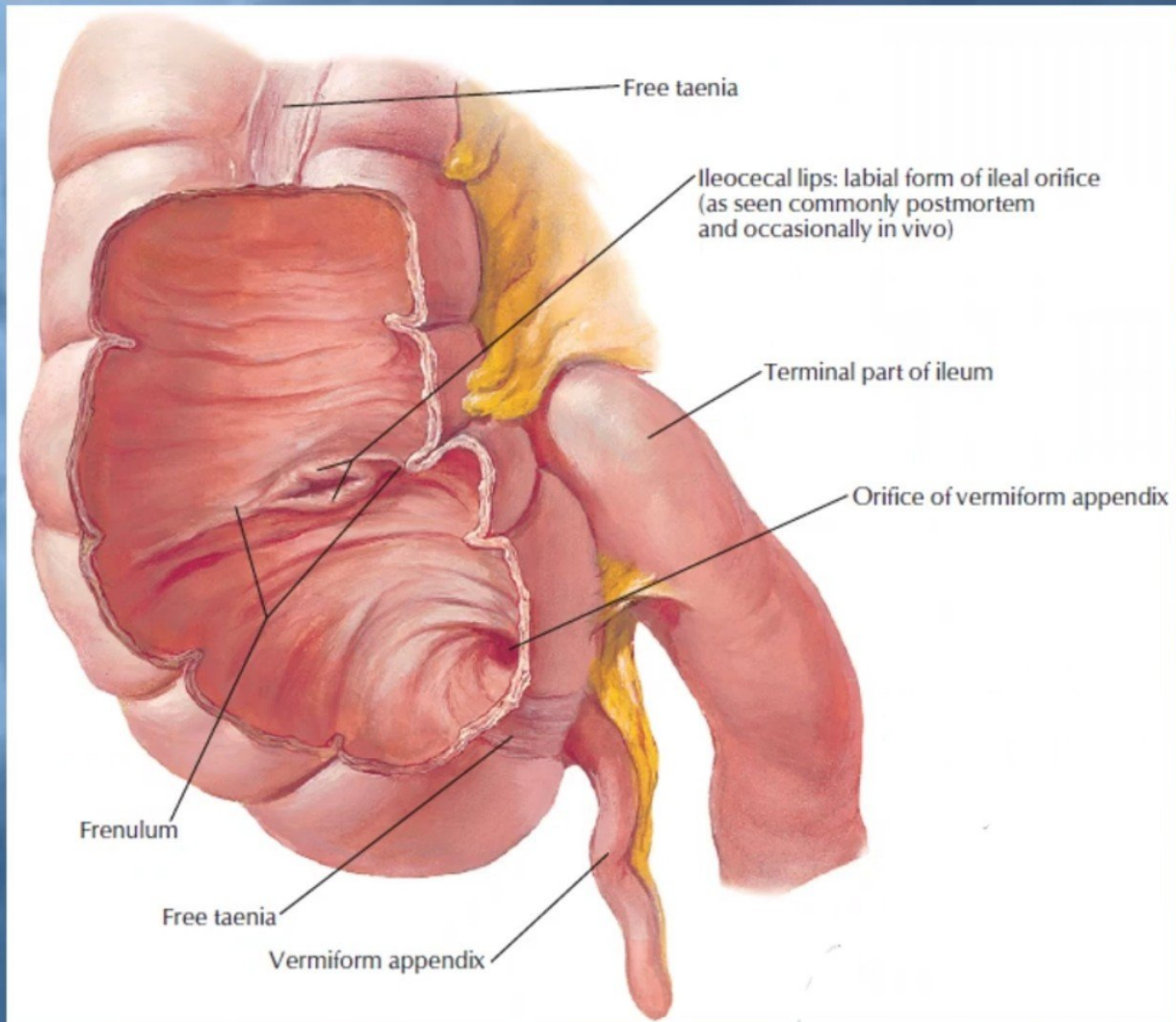
- ✓ It is a **blind-ended pouch**.
- ✓ It is situated in **the right iliac fossa**.
- ✓ It is **inferior** to ileocecal opening.
- ✓ It is about **2.5 in. (6 cm) long**.
- ✓ It is **completely covered with peritoneum (mobile)**.
- ✓ It **does not have a mesentery**.
- ✓ The **appendix** was attached to **posteromedial surface** of cecum.
- ✓ The presence of **peritoneal folds** in the cecum that creates the **superior ileocecal**, the **inferior ileocecal**, and the **retrocecal recesses**.



Cecum

Location and description inside:

- ✓ The terminal part of **the ileum** **enters the large intestine** at the junction of the cecum with the **ascending colon**.
- ✓ The opening is provided with **two folds**, or **lips**, which form the so-called **ileocecval valve**.
- ✓ The **appendix** communicates with the **cavity of the cecum** through an **opening** located below and behind the **ileocecval opening**.



Cecum Relations

Anteriorly:

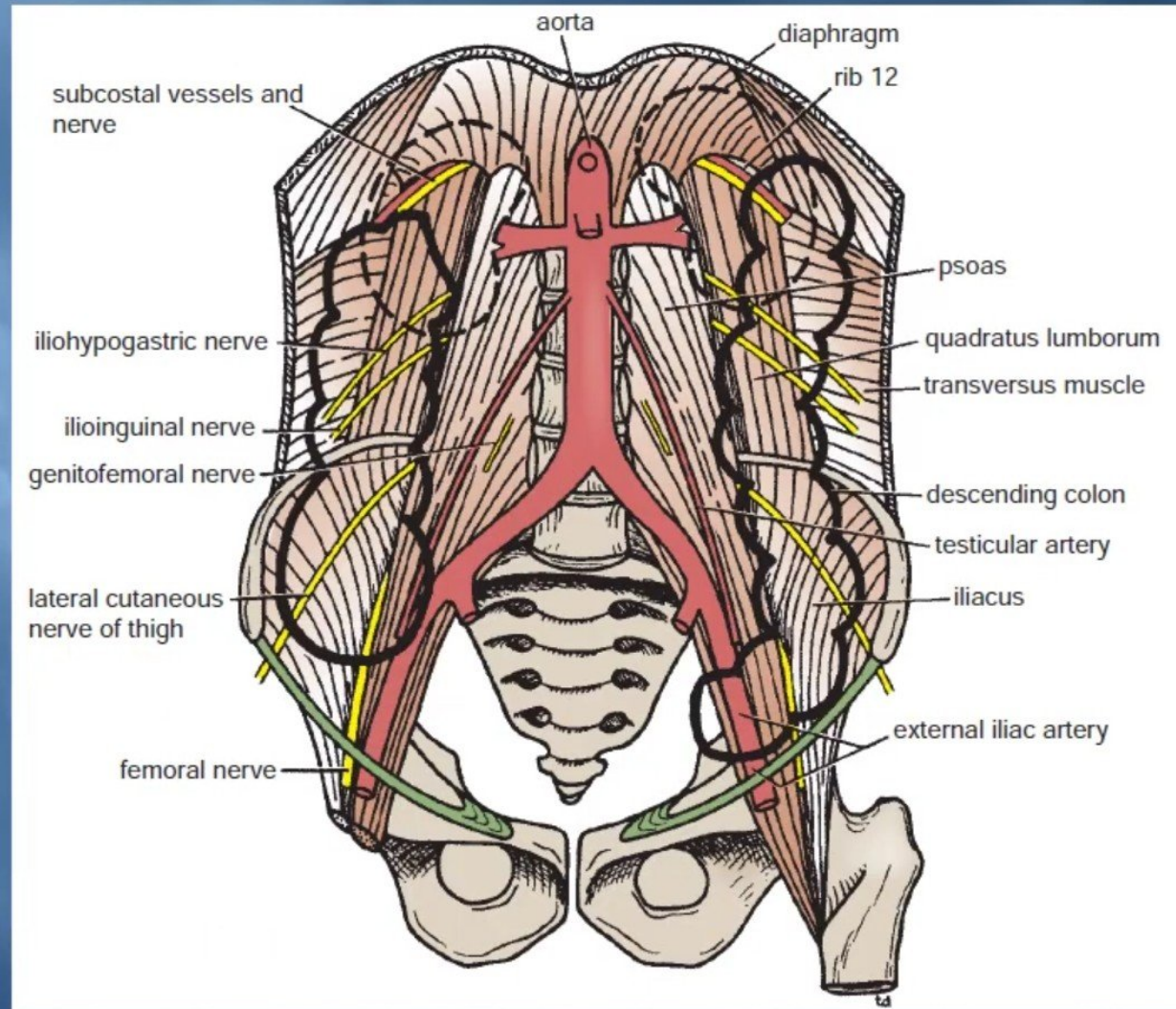
- ✓ The **Coils of small intestine**.
- ✓ The part of the **greater omentum**.
- ✓ The **anterior abdominal wall** in the **right iliac region**.

Posteriorly:

- ✓ The **psoas muscle**.
- ✓ The **iliacus muscle**.
- ✓ The **femoral nerve**.
- ✓ The **lateral cutaneous nerve of the thigh**.
- ✓ The **appendix** is commonly **found behind the cecum**.

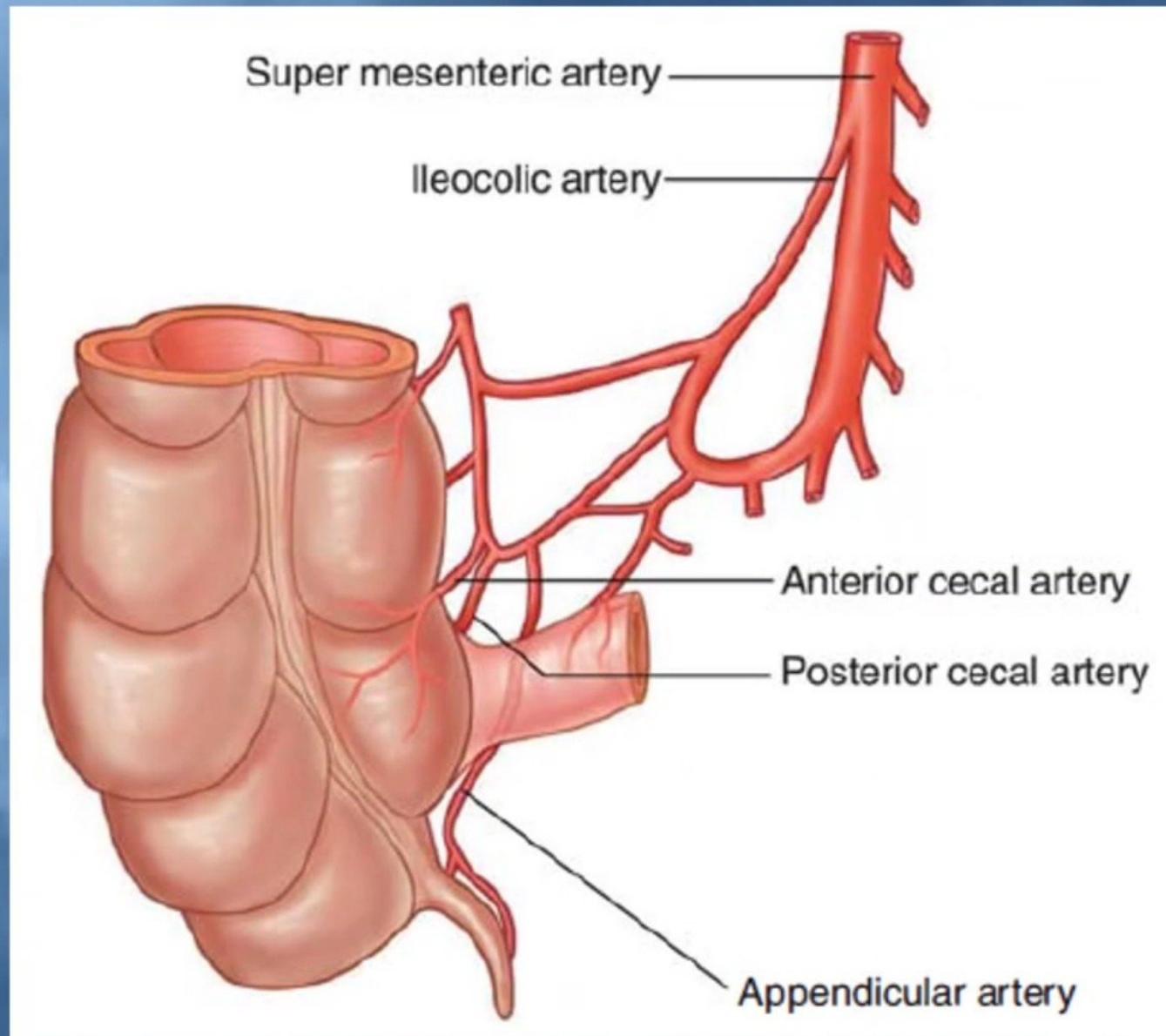
Medially:

- ✓ The **appendix** arises from **the cecum** on its medial side.



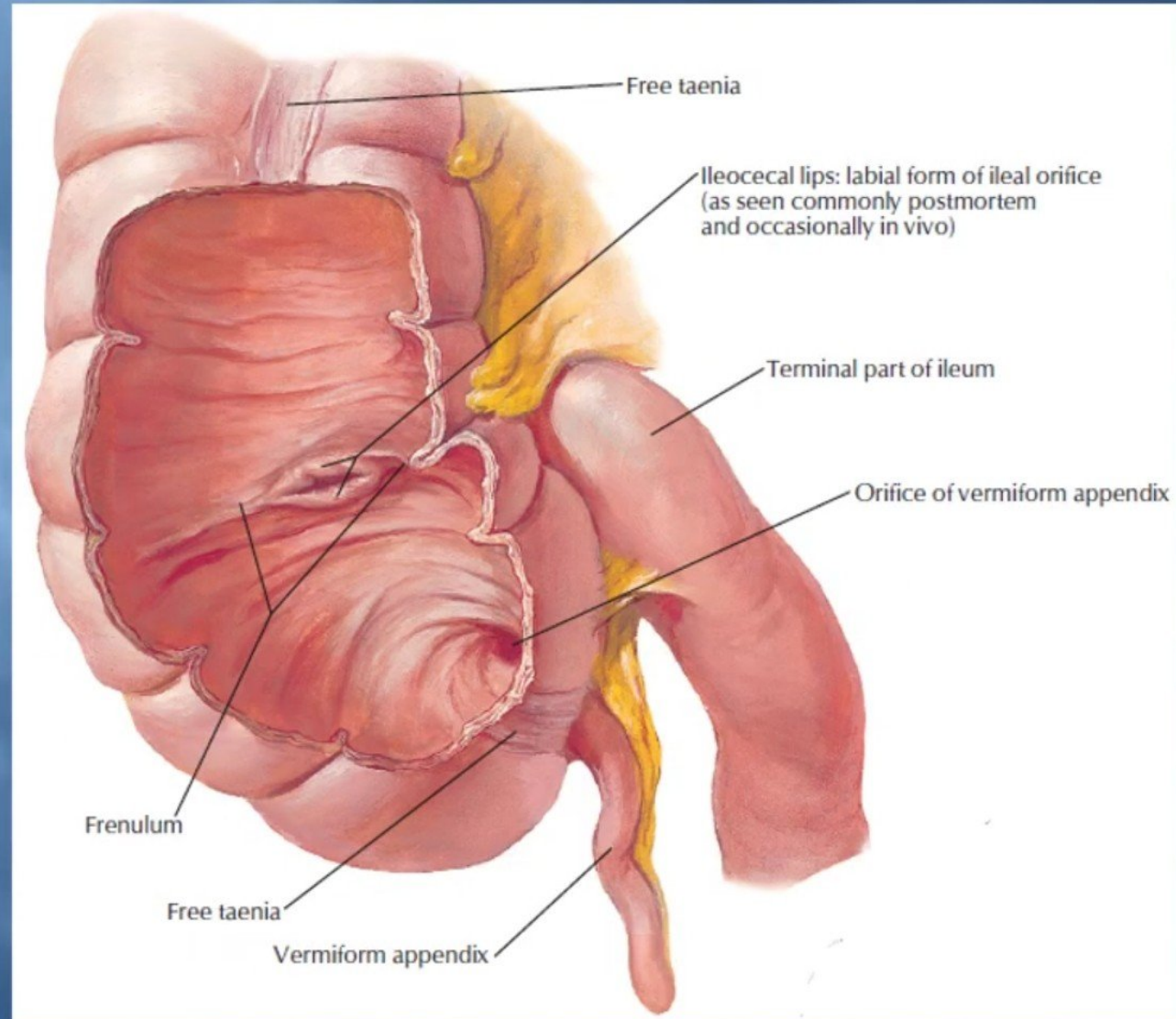
Cecum

- **Arterial supply:**
- ✓ **Anterior and posterior cecal arteries** form the **ileocolic artery**, a branch of the **superior mesenteric artery**
- **Veins:**
- ✓ The **veins correspond** to the arteries and drain into the **superior mesenteric vein**.
- **Lymph Drainage:**
- ✓ The lymph vessels pass through several **mesenteric nodes** and finally reach the **superior mesenteric nodes**.
- **Nerve Supply**
- ✓ Branches from the **sympathetic** and **parasympathetic (vagus) nerves** form the **superior mesenteric plexus**.



Ileocecal Valve

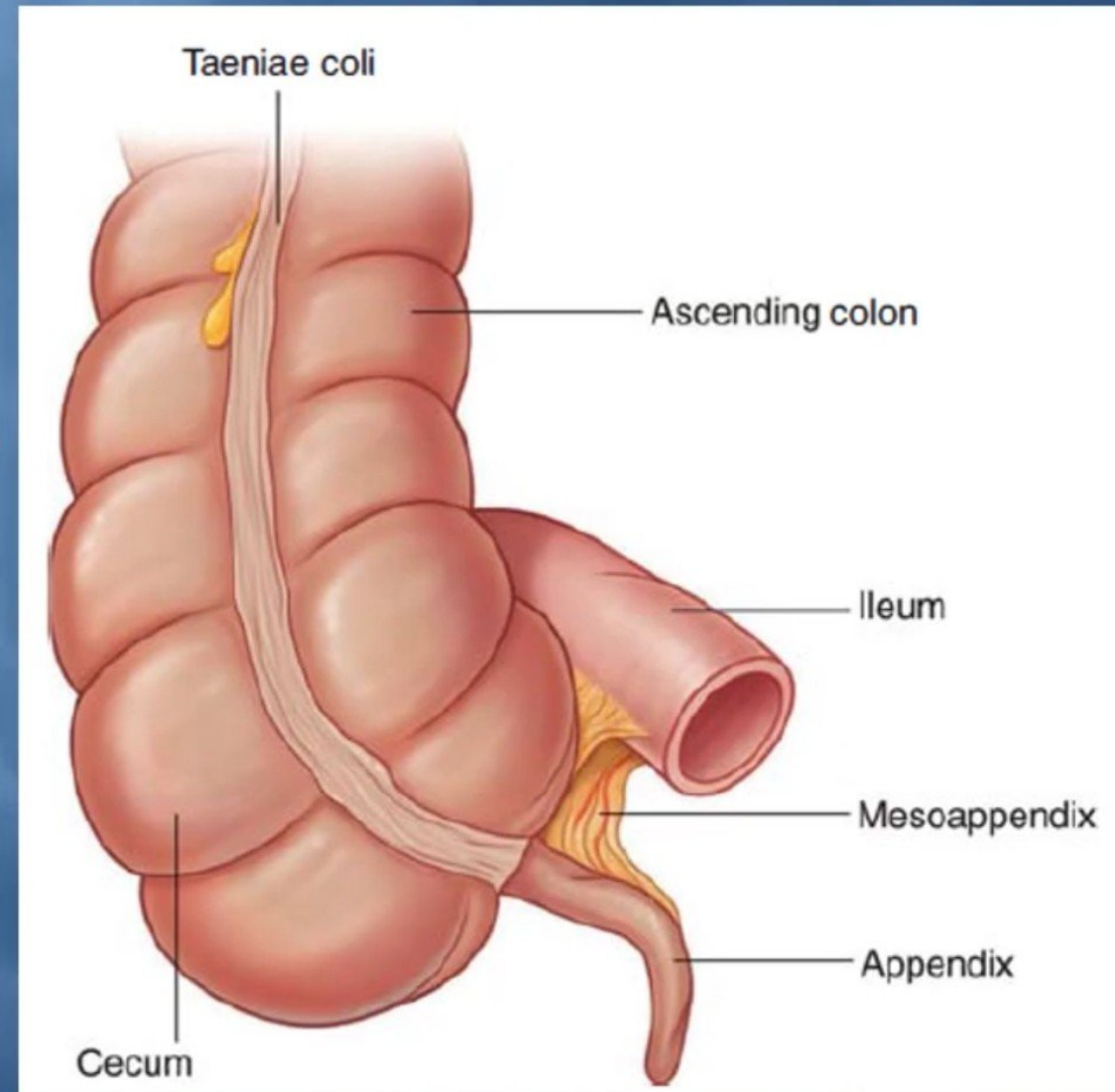
- A rudimentary structure.
- It consists of **two horizontal folds** of mucous membrane that project **around the orifice** of the ileum.
- The valve plays little or no part in the **prevention of reflux of cecal** contents into the ileum.
- The **circular muscle** of the lower end of the ileum (called the **ileocecocolic sphincter** by physiologists) serves as a sphincter and controls the flow of contents from **the ileum** into **the colon**.
- The **smooth muscle tone** is **reflexly increased** when the cecum is distended; the hormone **gastrin**, which is produced by the stomach, causes **relaxation of the muscle tone**.



Appendix

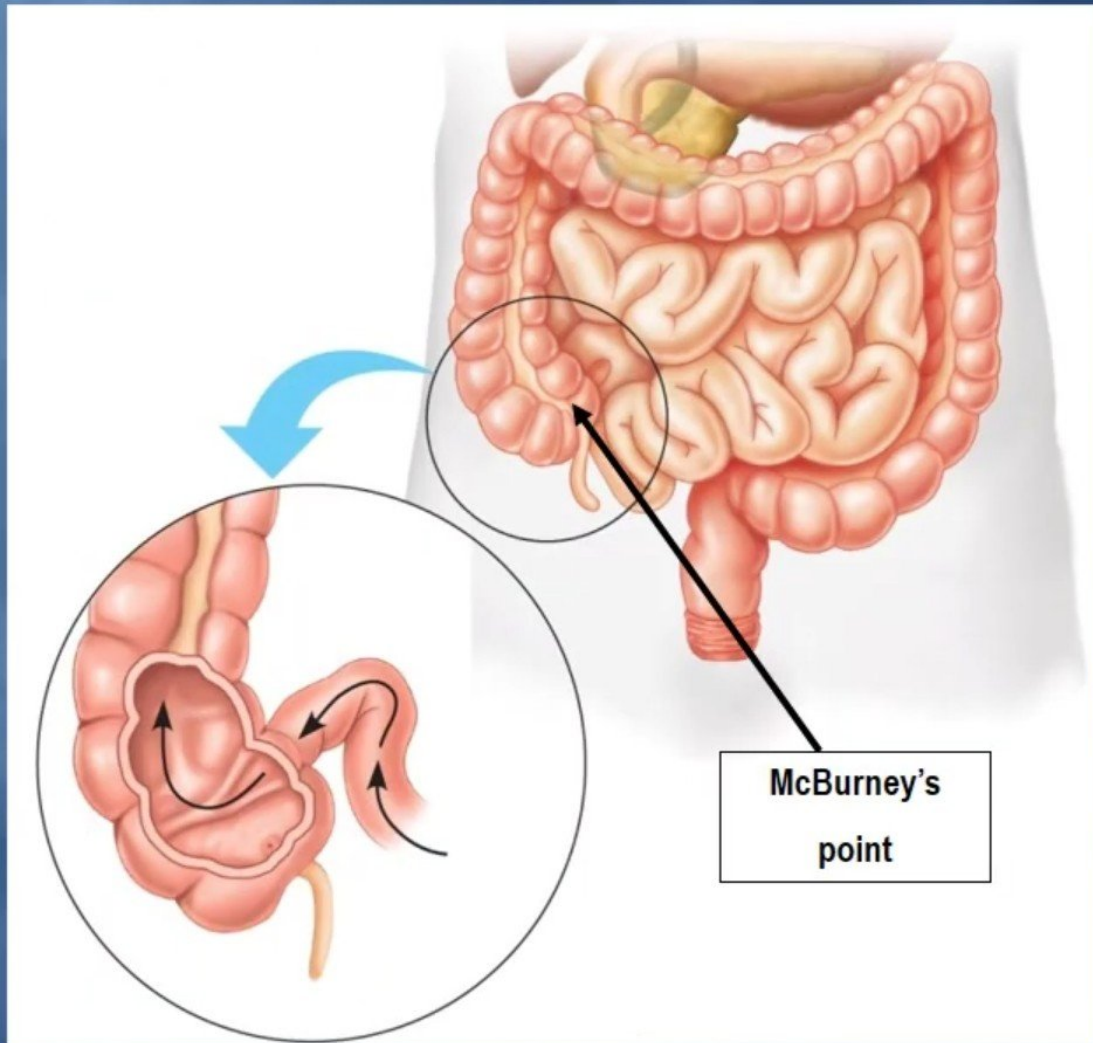
Location and Description

- It is a narrow, **muscular tube containing** a large amount of lymphoid tissue.
- It varies in **length from 3 to 5 in. (8 to 13 cm)**.
- The base is attached to the posteromedial surface of the cecum **about 1 in. (2.5 cm)** below **the ileocecal junction**.
- It has a **complete peritoneal covering**.
- It is attached to the mesentery of the small intestine by a short mesentery of (**the mesoappendix**).
- Among adult humans, the appendix is now thought to be involved primarily in immune functions.
- Appendix has great variations in position.



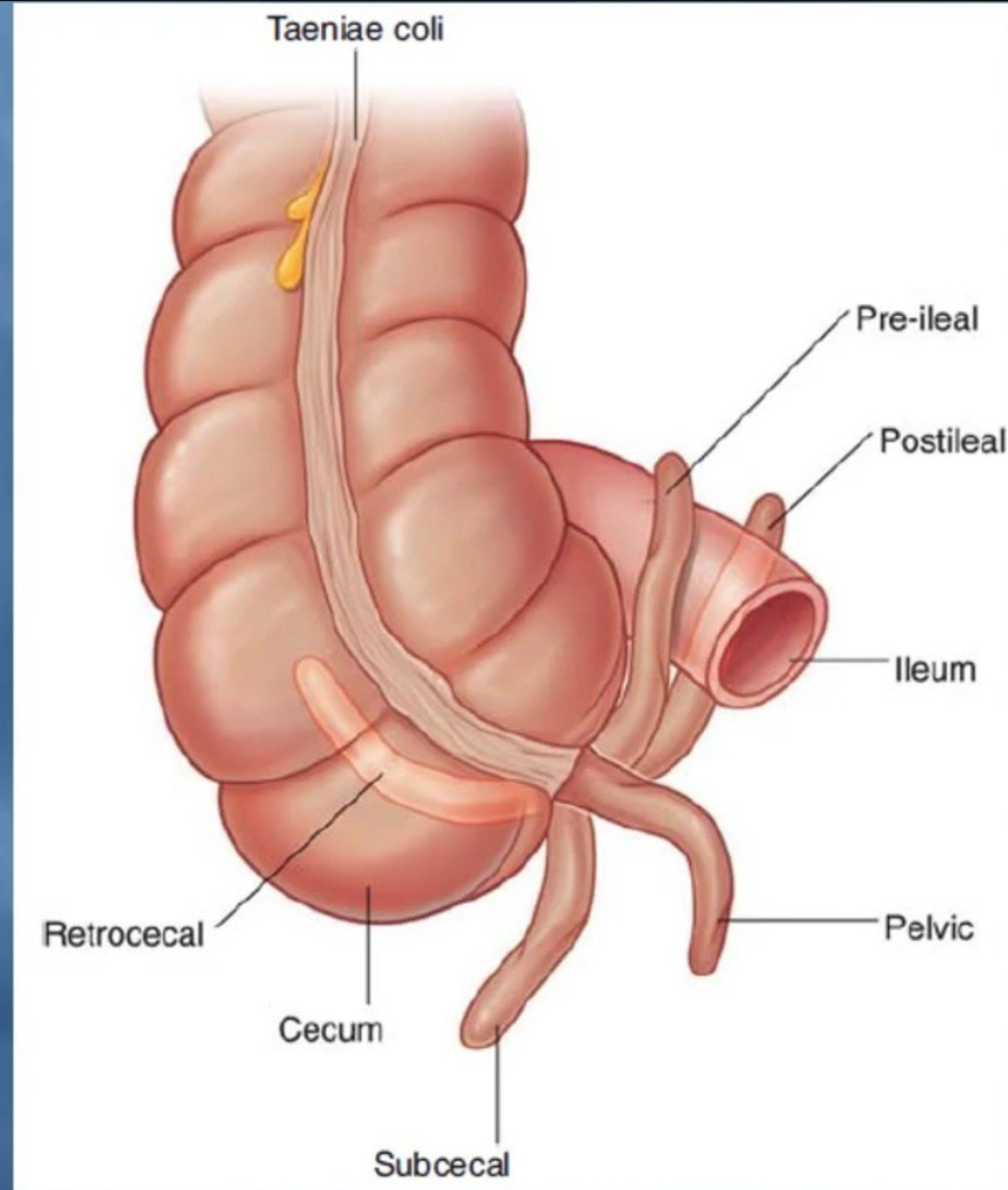
Appendix

- ✓ The **mesoappendix** contains the **appendicular vessels and nerves**.
- ✓ The appendix lies in **the right iliac fossa**, and in **relation to the anterior abdominal wall**.
- ✓ Its base is situated **one third** of the way up the line joining **the right anterior superior iliac spine** to **the umbilicus (McBurney's point)**.
- ✓ Inside the abdomen, the base of the appendix is easily found **by identifying the teniae coli of the cecum** and tracing them to the base of the appendix, where they converge to form a **continuous longitudinal muscle coat**.



Common Positions of the Tip of the Appendix

- **The tip of the appendix** is subject to a considerable range of movement and may be found in the following positions:
- (A) **Pelvic**: hanging down into the pelvis against the right pelvic wall.
- (B) **Retrocecal**: coiled up behind the cecum.
- (C) **Subcecal**: projecting upward along the lateral side of the cecum.
- (D) **Pre-ileal and post-ileal**: in front of or behind the terminal part of the ileum.
- **The first and second positions** are the most common sites.



The Appendix

Arteries

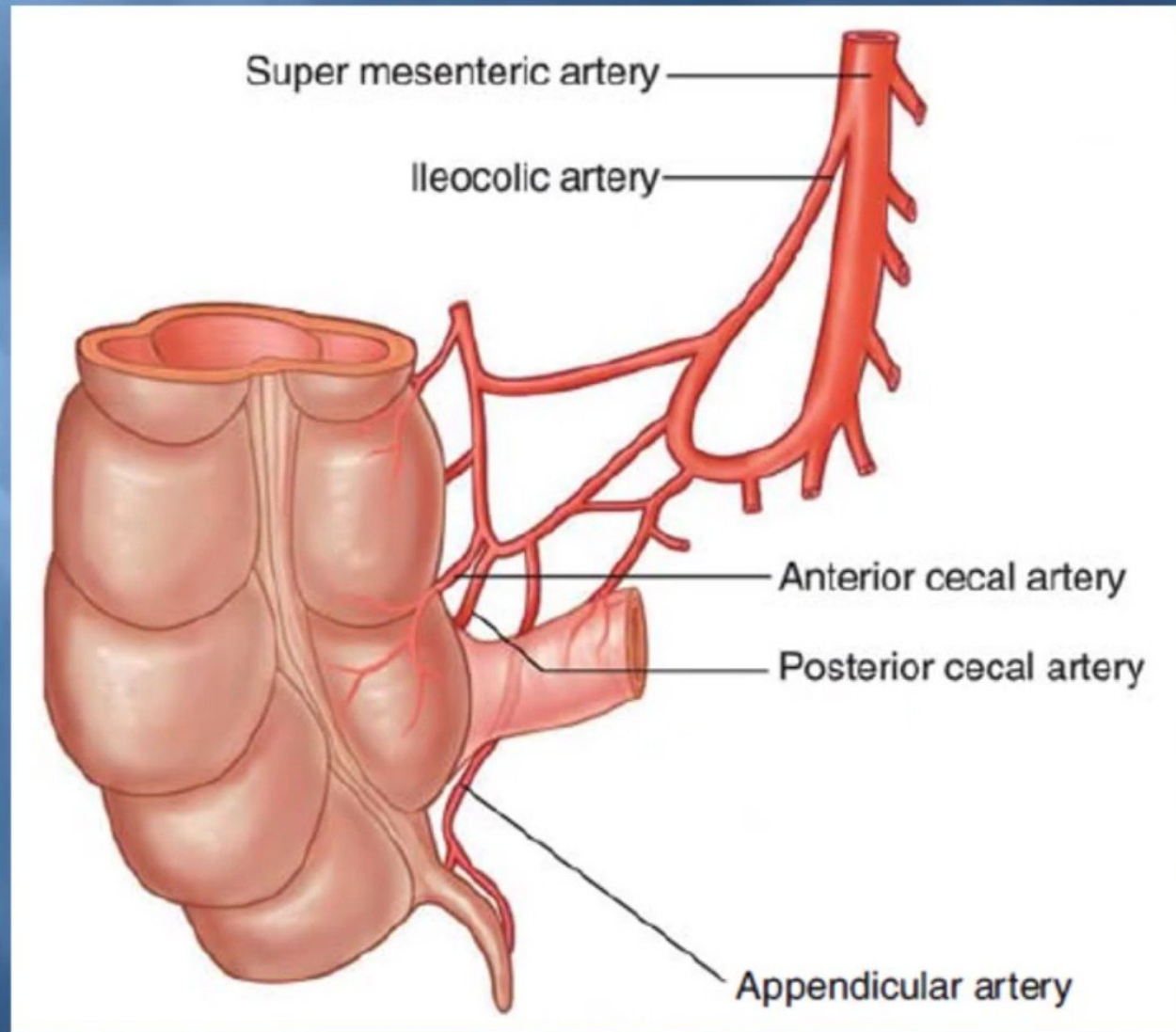
- ✓ The **appendicular artery** is a branch of the **posterior cecal artery**.

Veins

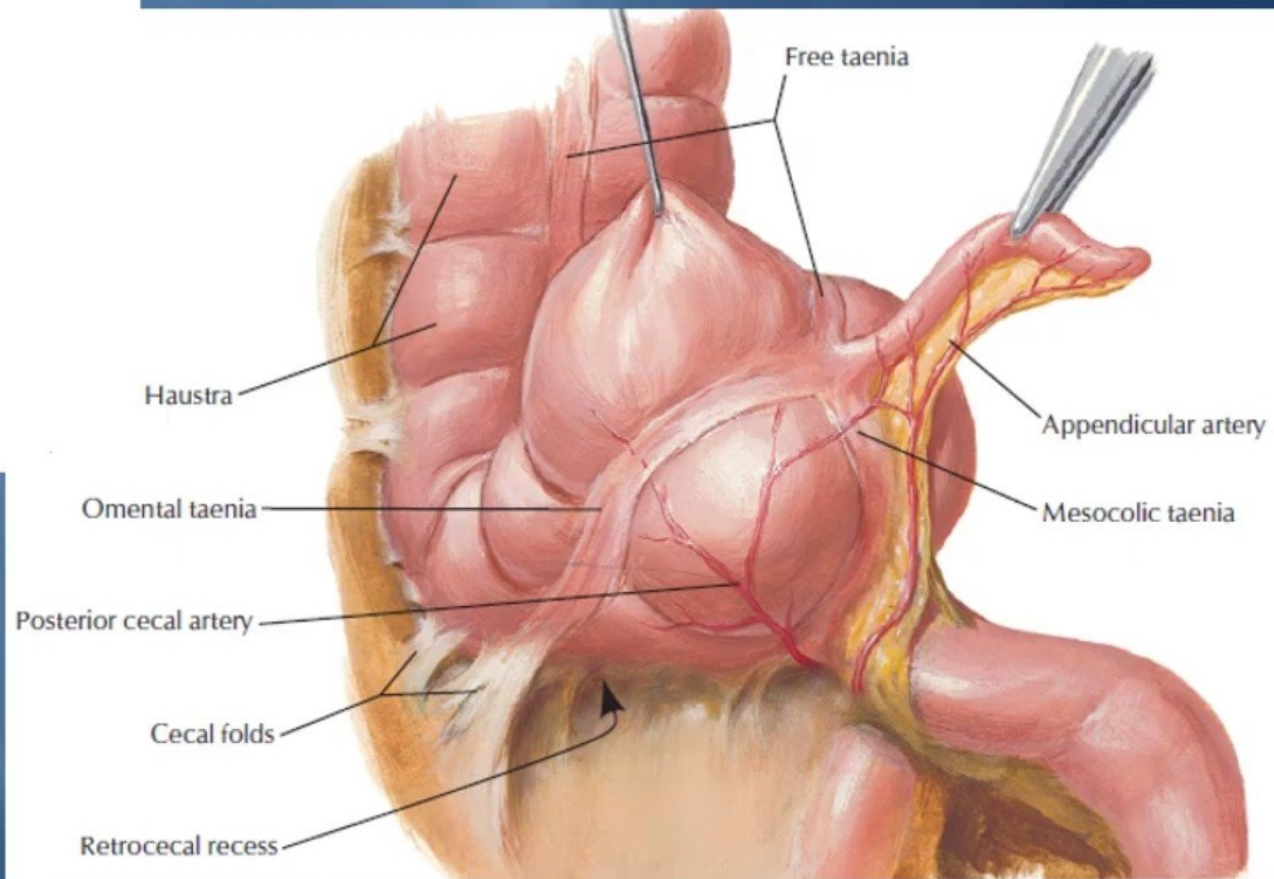
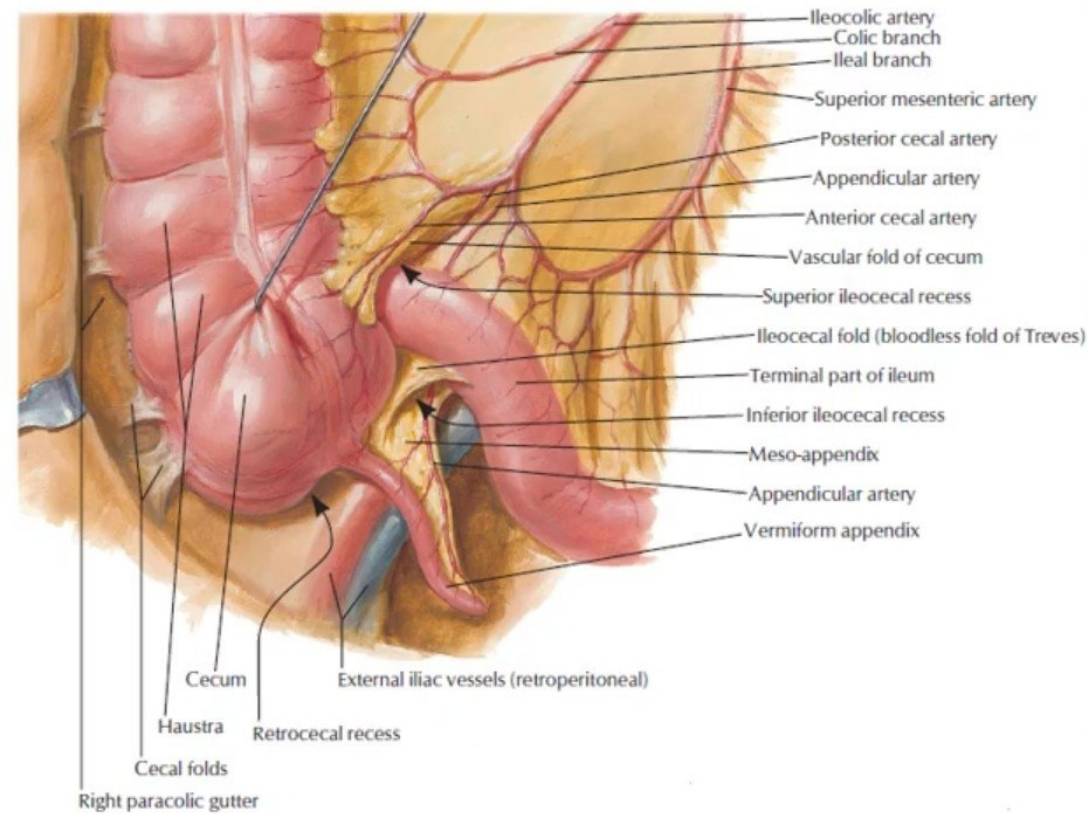
- ✓ The **appendicular vein** drains into the **posterior cecal vein**.

Lymph Drainage

- ✓ The **lymph vessels drain** into one or two nodes lying in the **mesoappendix** and then eventually into the **superior mesenteric nodes**.



The Appendix



Will be continue ...